

Md Shahriar Mannan Prottoy

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Professional Summary

Research-oriented Computer Science undergraduate with a strong interest in Artificial Intelligence, Deep Learning, and Computer Vision. Experienced in working on interdisciplinary research projects focused on healthcare and intelligent automation, with hands-on expertise in developing practical machine learning solutions. Familiar with collaborative research, problem-solving, and translating technical ideas into impactful real-world applications. Passionate about building reliable and interpretable AI systems while continuously exploring emerging technologies and research opportunities.

Work History

Vice President of Research and Innovation, *February 8, 2025—January 2026*
AR/VR & Game Development Lab, Daffodil International University, Dhaka, Bangladesh

- Professional and Academic Development
- Taken Sessions on Research Strategies & Writing
- Responsible for affordable AR implementation and Evaluation In Bangladesh Test Case

Education

Bachelor of Science *2022 – Present*
Daffodil International University, Bangladesh.
Major: Computer Science
Thesis Project: CNN-Tokenized Compact Transformer with Capsule Routing for Brain Tumor Detection.
CGPA: 3.95

Higher Secondary School Certificate *2019 – 2020*
Govt. Netrokona College, Bangladesh.
Major: Science
GPA: 5.00

Secondary School Certificate *2017 – 2018*
Anjuman Adarsha Govt High School, Bangladesh.
Major: Science
GPA: 4.89

Academic Achievements

- Regional Champion, NASA Space App Challenge(Global Nominee), 2025.
- Received Merit-Based Academic Waiver in all semesters for outstanding academic performance at Daffodil International University.
- Featured on the “Honor Board – Hero of the Hall” at Yunus Khan Scholar Garden-1 in recognition of academic excellence, discipline, and consistent performance.

Projects

- **Voice Of Bangladesh**-Led the development of a web-based platform as a Web Engineering Lab project to promote and share social awareness, public opinions, and national issues through an interactive online system.

- **Bank Management System**-Led a team in designing and developing a Java-based Object-Oriented Programming project featuring account management, transaction handling, and customer banking operations.
- **Hotel Management System**-Led the development of a Database Management System Lab project for managing hotel reservations, customer records, room allocation, and billing operations efficiently.
- **Food Delivery App**-Led the development of a Flutter-based mobile application as a Mobile Application Design Lab project, featuring local storage integration for managing food orders, user data, and delivery information.

Academic Activities & Competitions

- **Prompt Battle 2025** Participated in a competitive AI prompting event focused on creativity, problem-solving, and advanced prompt engineering techniques.
- **Unlock The Algorithm 2024** Competed in an algorithmic problem-solving contest emphasizing logical thinking, coding efficiency, and computational skills.
- **NASA Space App Challenge 2024, 2025** Participated in the global innovation hackathon organized by NASA, developing technology-driven solutions for real-world space and Earth challenges.
- **Take off 2022** Participated in a competitive programming contest organized to evaluate problem-solving ability and proficiency in C programming.
- **IOT FEST 2022** Participated in an IoT-focused technology festival showcasing innovative smart systems and embedded solutions.

Technical Skills

- **Programming Languages:** Python, C++, Html, CSS, Javascript, SQL
- **AI/ML Frameworks:** TensorFlow, PyTorch, Scikit-learn
- **Tools:** Jupyter Notebook, VS Code, LaTeX, Google Colab, Kaggle, Excel.
- **Natural Languages:** Bengali (Native speaker), English (IELTS Academic Overall Band: 7), Hindi (Listening and Speaking), Urdu (Listening and Speaking)

Research Publications

- Md Shahriar Mannan, Prottoy; Chowdhury , Mahtab ; Rahman, Redwan ; Tamim , Azim Ullah ; Rahman, Md Mizanur (2024), "PMRAM: Bangladeshi Brain Cancer - MRI Dataset ", Mendeley Data, V1, doi: 10.17632/m7w55sw88b.1
- M. S. M. Prottoy, M. Chowdhury, Y. A. Tahsin, M. H. H. Shanto, M. J. H. Sadi and A. Rahman, "Automated Lemon Leaf Disease Detection with Convolutional Neural Networks for Smart Agriculture," *2025 28th International Conference on Computer and Information Technology (ICCIT)*, Cox's Bazar, Bangladesh, 2025, pp. 5710-5715, doi: 10.1109/ICCIT68739.2025.11491793.
- S. S. Khan, M. S. M. Prottoy, M. Chowdhury, M. S. M. Siam, M. S. Hasan, and M. A. Rahman (2025). *Attention-Based Hybrid ViT-ResNet Model for Bangladeshi Mango Classification*. In *The 8th International Conference on Recent Trends in Image Processing & Pattern Recognition (RTIP2R)*
- S. S. Khan, M. S. M. Prottoy, M. Chowdhury, S. Sarder, M. J. H. Sadi, and M. A. Jisan, et al.(2025). *Lightweight Convolutional Networks and Explainable AI for Real-Time and Interpretable Tea Leaf Disease Detection*. In *IDAA:2025: International Conference on Intelligent Data Analysis and Applications*.

- M. S. Rahman, M. S. M. Prottoy, M. Chowdhury, T. T. Nidhi, A. U. Tamin, and S. S. Khan, et al. (2025). *CancerGuard: A Deep Learning Approach to Lung Cancer Detection*. In *IDAA:2025: International Conference on Intelligent Data Analysis and Applications*.
- A. Rahman, M. S. M. Prottoy, M. Chowdhury, M. H. H. Shanto, M. M. Haque, and S. Rahman, (2025) *Improving early Alzheimer disease diagnosis using machine learning and deep learning techniques*. In *IDAA:2025: International Conference on Intelligent Data Analysis and Applications*.
- A. Rahman, M. S. M. Prottoy, M. Chowdhury, Y. A. Tahsin, and S. Sarder, (2025). *Predicting cardiovascular disease risk using machine learning models: A comprehensive evaluation of clinical and ECG data analysis*. In *IDAA:2025: International Conference on Intelligent Data Analysis and Applications*.
- S. S. Khan, M. Chowdhury, M. S. M. Prottoy, K. Z. Haque, A. Momin, and T. A. Janisa, et al. (2025). *A Comprehensive Study on Deep Learning Architectures for Robust Object Identification in UAV-based Thermal Imaging*. In *IDAA:2025: International Conference on Intelligent Data Analysis and Applications*.

References

References available upon request.